

Jenna Melanson

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Education

University of British Columbia

Sept 2020 – Sept 2026

PhD in Zoology

Supervisor: Claire Kremen

Thesis: Navigating the risk-reward landscape: bumblebee movement, resource use, and health in agroecosystems

Massachusetts Institute of Technology

Sept 2016 – May 2020

Bachelor of Science in Biological Engineering

Minor: Literature

GPA: 5.0/5.0

Experience

Undergraduate Research Associate

Cambridge, MA

Parson's Laboratory — MIT

Sept 2019 – May 2020

Supervisor: Rachel Szabo, Otto Cordero (PI)

- Isolated microbial strains and sequenced gut microbiota from the Galapagos marine iguana to determine differences in microbial functional diversity between iguana subpopulations

Undergraduate Research Associate

Golden, CO

Department of Chemistry — Colorado School of Mines

May 2019 – August 2019

Supervisor: Anagha Krishnan, Matthew Posewitz (PI)

- Streamlined genomic DNA extraction protocols and characterized ploidy/post-transformational plasmid integration for *Picochlorum celeri*, an algal strain undergoing development as a biofuel source.

Field Technician

Glenwood, WA

Conboy Lake NWR — USFWS

May 2018 – August 2018

Supervisor: Trevor Sheffels

- Surveyed federally endangered/threatened species; performed invasive species removal and biocontrol in key habitat areas; directed Youth Conservation Corps (YCC) educational tours and biological data collection.

Undergraduate Research Associate

Cambridge, MA

Institute for Medical Engineering and Science / Broad Institute — MIT

March 2017 – May 2018

Supervisor: Brittany Goods, Alex Shalek (PI)

- Conducted longitudinal profiling of B cells across Ocrelizumab treatment in multiple sclerosis patients using single-cell RNA sequencing; performed bioinformatic analysis of single-cell RNA sequencing datasets.

Publications

Under Review

2. Melanson, J.B., Kelly, T.T., Clermont, N., Koch, J.B.U., Kremen, C. (submitted 07.2026). Agricultural intensification favours an introduced bumble bee over its native congener through differences in foraging range, habitat association, and lineage continuity. *Biological Conservation*.
bioRxiv: <https://doi.org/10.64898/2026.05.07.723627>
1. Arce-Valdés, L.R., Vilanova, J., Mahdjoub, H. Melanson, J.B., Kremen, C., Mlynarek, J.J., Khelifa, R. (submitted 04.2026). Diet composition across inter-sexual mimicry morphs: Do male-like females eat like males?

Published / Accepted

3. Knoerr, S., **Melanson, J.**, Carrillo, J., Kremen, C., Sargent, R. (accepted) Grassland set-asides provide nesting habitat for bumble bees in agroecosystems. *Journal of Applied Ecology*.
2. **Melanson, J.B.**, Kelly, T.T., Waters, S.M., Kremen, C., & Ponisio, L. C. (2026). Agricultural intensification shapes community composition and parasite prevalence of native but not introduced bumble bees. *Basic and Applied Ecology*, 96, 151836. <https://doi.org/10.1016/j.baae.2026.06.008>
1. Krishnan, A., Likhogrud, M., Cano, M., Edmundson, S., **Melanson, J. B.**, Huesemann, M., McGowen, J., Weissman, J. C., & Posewitz, M. C. (2021). Picochlorum celeri as a model system for robust outdoor algal growth in seawater. *Scientific Reports*, 11(1), 11649. <https://doi.org/10.1038/s41598-021-91106-5>

Talks and Posters

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- 2026
- Investigating pollinator foraging behaviour with spatially explicit capture-recapture models. StanCon, Uppsala, Sweden, August 2026.
- 2025
- Introduced bumblebee forages further and exhibits different landscape ecology than a native congener. British Ecological Society, Edinburgh, UK, December 2025.
- Effects of invasive *Bombus impatiens* on native bumble bee disease prevalence are parasite-specific. Pacific Branch Entomological Society Conference, Salt Lake City, Utah, USA, April 2025.
- 2024
- Bumble bee disease dynamics in agroecosystems: The roles of landscape-scale management and an introduced species. Entomology, Phoenix, Arizona, USA, November 2024.
- Bumble bee parasite prevalence in the Lower Fraser Valley: the impacts of *B. impatiens* abundance on native *Bombus spp.* North American *B. impatiens* Working Group Meeting, Presented Virtually, August 2024.
- Bumble bee parasite prevalence in agroecosystems of the Lower Fraser Valley. Canadian Society for Ecology and Evolution (CSEE), Vancouver, BC, Canada, May 2024.
- 2022
- Poster:* Melanson, J.B., Kelly, T., Kremen, C. Navigating the risk-reward landscape of agroecosystems: bumble bee movement and resource use across a landscape-diversity gradient. EcoEvo, Squamish, BC, Canada, October 2022. Received Student Presentation Best Poster Prize.

Awards

Peter Wall Graduate Research Award (USD 22,000)	2025-2026
Vanier Canada Graduate Scholarship (USD 110,000)	2022-2025
UBC Four-Year Fellowship (USD 53,000)	2020-2024
National MS Society Scholarship (USD 12,000)	2016-2020
Hadden Youth Foundation Scholarship (USD 16,000)	2016-2020

Service and Outreach

Biology Undergraduate Diversity in Research Program — Provided professional development training to a total of six mentees. Provided lab experience/training for one BUDR student.	2025, 2022-2023
Disabled in STEM Mentorship Program — Provided lateral mentorship and served as a group leader in the 2025 cohort, aimed at providing academic supports and improving accessibility for students and scientists pursuing careers in STEM.	2025
Debbie & Justin Wragg-Schmidt Zoology Symposium Academic Committee — Solicited, organized, and scheduled academic presentations for the UBC Department of Zoology's annual graduate student symposium.	2025
UBC Multidisciplinary Undergraduate Research Conference Adjudica-	2025

tor — Provided feedback on undergraduate research projects presented at the annual conference.

Future of Food Global Dialogue and Webinar Series — Convened diverse panelists (including representatives from academia, government, industry, and the local farming community) for webinar, “Unlocking the Potential of Diversified Farming: Exploring Benefits and Overcoming Barriers.” 2023

Outreach to Contributing Landowners — Provided documentation on floral / bumble bee diversity and abundance to 35+ landowners who participated in thesis data collection 2023

Let’s Talk Science Program Lessons in Ecology and Evolution Fundamentals (LEEF) — Gave lesson on pollinator biodiversity to elementary students in Vancouver 2022

Little Beavers MIT — Co-founded an MIT chapter of Step Ahead, a national non-profit that collaborates with collegiate athletes to deliver free running clubs for kids with autism. 2019-2020

MIT Student Athlete Advisory Committee — Provided input and review on policies governing NCAA student-athletes 2017-2020

Teaching Experience

Fundamentals of Biostatistics (UBC BIOL 300) — *Professors Kaitlyn Gaynor, Vikram Balega* 2025

- Led two weekly lab sections of 25 students each to develop skills in coding and statistical hypothesis testing in R; provided office hours to improve student understanding of statistical theory; designed review questions; graded assignments and exams

Biology of the Cell (UBC BIOL 112) — *Dr. Karen Smith* 2022

- Led two weekly tutorial sections of 50 students each to revisit key learning goals, designed review questions, graded assignments and exams

Molecular Genetics (UBC BIOL 336) — *Professor Craig Berezowsky* 2021

- Led two weekly sections of 50 students each, facilitated group problem solving, developed/modified exam questions, graded assignments and exams

Mentorship

Study design / statistical analyses 1 undergraduate (directed studies co-advisor)

- e.g., determining appropriate sample size and distribution of survey effort

Molecular biology protocols 1 postdoc, 1 PhD candidate, 5 undergraduates

- e.g., DNA extractions, PCR/multiplex PCR, gel electrophoresis, molecular metabarcoding

Field data collection 3 undergraduates

- e.g., bumble bee ID, pollinator surveys, flowering vegetation surveys

Database development and management 4 undergraduates

- e.g., database design/construction, data entry and cleaning

Non-Academic Awards & Experience

2024 U.S. Olympic Team Trials Qualifier, Women’s Track and Field

2019 NCAA Elite 90 Recipient, Women’s Track and Field